J Mater Sci: Mater Electron **(2025) 36:824**

Comprehensive study on fabrication, characterization and performance evaluation of Ag/TiO_x/ITO and Au/Ag/TiO_x/ITO memristors

Kubra Saka[1,2], Dincer Gokcen[1,*], Halil Ibrahim Efkere[3], Cem Bayram[4], and Suleyman Ozcelik[3,5]

[1] *Department of Electrical and Electronics Engineering, Graduate School of Science and Engineering, Faculty of Engineering, Hacettepe University, Ankara, Turkey*
[2] *Department of Electrical and Electronics Engineering, Erzurum Technical University, Erzurum, Turkey*
[3] *Photonics Application and Research Center, Gazi University, Ankara, Turkey*
[4] *Department of Nanotechnology and Nanomedicine, Graduate School of Science and Engineering, Hacettepe University, Ankara, Turkey*
[5] *Department of Photonics, Faculty of Applied Sciences, Gazi University, Ankara, Turkey*

Received: 3 March 2025
Accepted: 27 April 2025
Published online: 18 May 2025

© The Author(s), 2025

ABSTRACT

The structure of electrodes and active materials in memristors plays a critical role in determining their electrical behavior. This study primarily focuses on Ag/TiO_x/ITO and Au/Ag/TiO_x/ITO memristors incorporating titanium oxide (TiO_x) thin films with varying thicknesses (25, 50, and 100 nm), deposited via RF magnetron sputtering onto ITO-coated glass substrates. Comprehensive surface characterization techniques, including scanning electron microscopy (SEM), electron dispersive X-ray spectroscopy (EDX), atomic force microscopy (AFM), Raman spectroscopy, and X-ray photoelectron spectroscopy (XPS), were employed to examine the morphological, structural, and chemical properties of the films. SEM–EDX mapping revealed thickness dependent elemental distributions. AFM analysis showed increased surface roughness (1.69 to 2.15 nm) and grain size (60 to 90 nm) with increasing film thickness. XPS measurements conducted at 8-week intervals provided insights into the oxidation behavior of Ag electrodes and their surface chemistry evolution over time. Raman analysis confirmed the presence of anatase and rutile phases in the titanium oxide thin films, as evidenced by the characteristic peaks observed at 196.83 cm^{-1} and 607.43 cm^{-1}, respectively. Focused ion beam (FIB) and scanning transmission electron microscopy (STEM)-EDX were also utilized to investigate elemental boundaries between the layers. Bipolar resistive switching behavior, without the need for any electroforming process, was observed in all devices within 7 days of fabrication and remained stable throughout 1000 current–voltage (I–V) cycles. After 8 weeks of storage, further endurance tests were conducted using ± 1.5 V SET and RESET voltages, with HRS and LRS values measured at a read voltage of 0.5 V. Notably, the highest HRS/LRS ratio of

Address correspondence to E-mail: dgokcen@hacettepe.edu.tr

https://doi.org/10.1007/s10854-025-14894-w

16.07 was achieved in the Ag/TiO_x/ITO memristor with a 100 nm TiO_x layer. These findings underscore the critical influence of oxide thickness on resistive switching performance and endurance. Furthermore, the comparative evaluation of Ag and Au-passivated Ag electrodes highlights the significance of electrode configuration in enhancing memristor reliability and long-term operational stability.

1 Introduction

Hardware mimicking the biological nervous system, known as neuromorphic devices, offers innovative solutions to modern computing challenges. As the limits of traditional digital computing are approached, the demand for novel devices and materials is rising. Device engineering-based approaches aim to emulate the human brain's efficient and parallel processing capabilities. In this context, memristors play a crucial role in mimicking synaptic functions in neuromorphic hardware. Memristors (memory-resistors) are passive and two-terminal circuit elements that establish a direct relationship between electrical charge and magnetic flux [1, 2]. Memristors can dynamically adjust resistance based on the applied voltage and maintain their states (non-volatile memory). The novel properties of memristors make them highly suitable for computing and data storage applications. Owing to their synaptic behavior, memristors support learning and memory processes in hardware akin to biological networks [3, 4]. This functionality has the potential to offer unique properties with reduced energy consumption in the hardware-level design of artificial neural networks and parallel data processors [5–8].

The first study on resistive switching used silicon dioxide as the insulating material (active layer) [9], followed by improvements using various oxides, titanium oxide [10–13], zinc oxide [14, 15], copper oxide [16, 17], hafnium oxide [18, 19], silicon oxide [20, 21], and nickel oxide [22, 23]. The active layer affects key parameters, including switching mechanism, speed, conductivity, and resistance ratios. Structural changes occur in the active layer according to the applied voltage and current, and these changes reveal bow-tie hysteresis in I-V characteristics. Electrode materials are as important as the active layer because they interact with the active material and directly affect stability, performance, and endurance. Mostly investigated electrode materials include copper [24, 25], silver [26], platinum [27, 28], nickel [29, 30], gold [31], aluminum [32], indium tin oxide [33], and graphene [34, 35].

Resistive switching behavior depends on electrode and active layer materials, where an electric field induces ion migration in metal–insulator–metal structures, forming conductive filaments between electrodes. This phenomenon is referred to as filamentary-type switching. Two primary mechanisms are defined based on the origin of filament formation. Electrochemical metallization memory (ECM), involves the migration and reduction of metal cations to form metallic filaments [36, 37]. Valence change memory (VCM) relies on the movement of oxygen vacancies within the oxide to generate conductive paths [38, 39]. In contrast, some devices exhibit interface-type switching, where the electrical characteristics are governed by the nature of the contact at the electrode/active layer interface. In such cases, the contact type, whether ohmic or Schottky, is crucial in determining the overall device performance.

Since the advent of the memristor, titanium oxide has been extensively studied as an active layer in memristor structures. Our study uses sputtering technique to focus on the memristive behaviors and performance of titanium oxide thin films with varying thicknesses (25, 50, and 100 nm) deposited on ITO-coated glass substrates. It is critical to note that surface characterization techniques are covered in detail to understand the relationship between thin film properties, electrical behavior, and durability. The configurations examined included Ag/TiO_x/ITO and Au/Ag/TiO_x/ITO structures. Existing literature highlights the relationship between fabrication methods and the thicknesses of titanium oxide layers in memristor devices employing Ag top and ITO bottom electrodes [40–42]. For instance, Kavehei et al. demonstrated that TiO_2 thin films fabricated via the atomic layer deposition method exhibit bipolar memristive behavior in

$Ag/TiO_2/ITO$ configuration, requiring an electroforming process [40]. Their results showed a high resistance state (HRS) to low resistance state (LRS) ratio of approximately 55, sustained over 100 cycles. Similarly, Elshekh et al. investigated the memristive properties of 400 nm thick titanium oxide thin films deposited via sputtering in $Ag/TiO_x/ITO$ configuration [41]. Their analysis included energy-dispersive X-ray spectroscopy and scanning electron microscopy (SEM), and electrical measurements presented an HRS/LRS ratio of 2.9 after endurance tests over 75 cycles. Park et al. fabricated $Ag/TiO_2/TiO_x/ITO$ memristor structures using spin-coating and reported their operations within a relatively low voltage range (− 1 V to 1 V) [42]. Their endurance tests with 10 cycles demonstrated an HRS/LRS ratio of 2×10^3. Structural and compositional analyses were done using high-resolution transmission electron microscopy (TEM) and X-ray photoelectron spectroscopy (XPS).

The present study uses sputtering technique to examine the memristive behavior and performance of titanium oxide (TiO_x) thin films deposited at varying thicknesses on ITO-coated glass substrates. Ag and Au/Ag thin films were used as top electrodes. The most important aim of the study is to expand the potential applications of titanium oxide-based memristors by providing a comprehensive perspective on thin film characterization and performance evaluation. The surface characterizations of TiO_x thin films were carried out using scanning electron microscopy (SEM) and atomic force microscopy (AFM). Focused ion beam (FIB) and scanning transmission electron microscopy (STEM) with EDX were used to analyze interlayer diffusion within $Ag/TiO_x/ITO$. The memristive behavior of the fabricated structures, and their stability, were evaluated using a two-terminal current–voltage measurement system. Furthermore, endurance tests were conducted over 1000 cycles on $Ag/TiO_x/ITO$ and $Au/Ag/TiO_x/ITO$ structures with titanium oxide active layers of different thicknesses. The highest resistance ratio was observed in the $Ag/TiO_x/ITO$ structure with a 100 nm thick active layer. According to the endurance tests, Ag/TiO_x(100 nm)/ITO structure demonstrated higher performance and the highest resistance ratio than the gold-passivated silver (Au/Ag) electrodes. While Au/Ag electrodes also showed promising performance, the resistance ratios achieved by these structures were more limited than those observed with Ag electrodes. While titanium oxide-based memristive devices have been widely investigated, most previous studies have primarily focused on electrical performance without establishing a clear correlation between film thickness and structural and switching characteristics. In contrast, the present study employs a multi-dimensional approach to provide a comprehensive and systematic analysis of the influence of materials and film thicknesses on device performance. Integration of advanced characterization techniques such as SEM–EDX, AFM, FIB, STEM-EDX, XPS, and Raman spectroscopy is an approach rarely combined in a single study. Moreover, including long-term endurance after 8 weeks offers novel insights into the temporal stability and reliability of the memristor structures.

2 Materials and methods

TiO_x coatings with different thicknesses (25, 50, 100 nm) were deposited on ITO-coated glass (185 nm, resistivity 10 Ω/sq), Si wafer, and glass substrates by using the planar RF magnetron sputtering technique. Before the deposition process, substrates were sequentially cleaned in acetone and alcohol for 5 min each in an ultrasonic bath. Subsequently, all cleaned substrates were dried with high-purity nitrogen gas and placed in the loading chamber of the Bestec Sputtering System under vacuum. The Bestec Sputtering System consists of two vacuum chambers: a loading chamber that can accommodate up to 10 different sample holders simultaneously and a main chamber where the deposition processes are carried out. Loading multiple sample holders in the loading chamber at once is a significant advantage for performing sequential, similar, and comparative depositions. This is because the samples are kept in the same vacuum atmosphere from the beginning of the first deposition until the end of the last deposition. Once the vacuum level of the loading chamber reached a value equivalent to that of the main chamber, the samples were transferred to the main chamber for TiO_x deposition. Titanium oxide thin films were deposited using a 99.99% pure titanium oxide target. The deposition parameters used for the 25, 50, and 100 nm coatings were as follows: base vacuum of 10^{-8} mbar, power of 150 Watts, gas flow rate of 10 sccm (Ar), and target-to-substrate distance of 90 mm. After the deposition of TiO_x thin films, the film thickness was determined using a Veeco Dektak 150 Stylus-Type Profilometer. During the sputtering process, the film thickness was controlled by varying

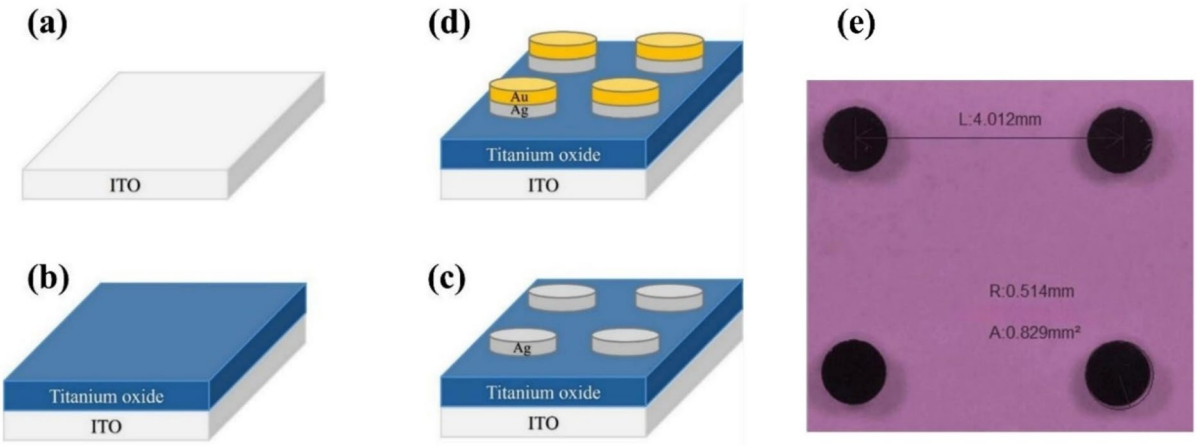


Fig. 1 Deposition process flow of titanium oxide-based sandwich-type memristors; **a** ITO coated glass substrate, **b** deposition of titanium oxide layer using sputtering, deposition of **c** Ag (150 nm) top electrode and **d** Au (15 nm) using thermal evaporation method, and **e** Representative top view image of Ag/TiO_x/ITO memristors (Here, R represents the radius of the electrode; A denotes the surface area of the electrode; and L corresponds to the distance between the two electrodes)

the deposition time while keeping other parameters constant. Accordingly, titanium oxide films with thicknesses of 25 nm, 50 nm, and 100 nm were obtained at deposition durations of 11, 24, and 55 min, respectively. The surface properties of the TiO_x thin films produced with different thicknesses were characterized using SEM–EDX (Tescan GAIA with Oxford XMAX 150 EDS) and AFM (Nanomagnetics ezAFM) techniques, the structural properties were determined using Raman spectroscopy (B&W TEK InnoRam), and the stoichiometric analyses were performed using XPS (Thermo Fisher Scientific K-Alpha). To investigate the memristor properties of the thin films obtained after the depositions, Ag (150 nm) and Ag (150 nm)/Au (15 nm) top electrodes were deposited using a thermal evaporation system with 1 mm diameter shadow masks. For the silver deposition, evaporation power of 280 W was applied for 18 min, resulting in a film thickness of 150 nm (Fig. 1). For 15 nm thick gold deposition, evaporation power of 310 W was applied for 4 min. After completing the contact structures, Ag/TiO_x/ITO and Au/Ag/TiO_x/ITO devices were analyzed for their memristor properties using a two-probe sourcemeter. The fabrication process has been simplified by using ITO as both the bottom electrode (conductive layer) and the substrate.

The long-term stability of fabricated memristor structures was examined based on the active layer thickness and the top electrode. Two types of memristor structures Ag/TiO_x/ITO and Au/Ag/TiO_x/ITO were produced to investigate the passivation effect of the gold layer on the silver top electrode. After depositing the contact electrodes, the first XPS measurement was conducted on the silver surface (not between the silver and the active layer). The samples were then stored in a desiccator (Secador). Current–voltage measurements obtained 7 days after fabrication revealed forming-free bipolar resistive switching behavior over 1000 (I-V sweep) cycles. Eight weeks later, second XPS measurements were taken from the silver contact surface to monitor their oxidation tendencies and chemical changes, providing insights into the long-term performance of the memristor structures. Endurance tests were conducted for 1000 cycles to determine whether the oxidation of the silver top electrode affected the performance of the structures over time. The endurance test results of the Ag/TiO_x/ITO and Au/Ag/TiO_x/ITO memristor structures demonstrated the impact of varying titanium oxide layer thicknesses on memristive performance. The HRS/LRS ratios of the Ag/TiO_x/ITO and Ag/TiO_x/ITO structures were calculated, where the highest ratio of 16.07 was observed for the Ag/TiO_x(100 nm)/ITO structure. Additionally, FIB and STEM-EDX analyses performed after endurance tests detailed the elemental distribution and interlayer diffusion within the Ag/TiO_x/ITO memristor structure. Notably, significant diffusion traces were not observed at the boundary between the Ag

electrode and the TiO_x active layer. Endurance tests conducted after 8 weeks further validated the long-term stability of the structures.

Steps followed for the fabrication and characterization of titanium oxide-based memristors:

- Titanium oxide (TiO_x) layers (25, 50, and 100 nm) were deposited onto cleaned ITO substrates using the planar RF magnetron sputtering technique.
- SEM–EDX was utilized for elemental analysis.
- AFM was employed to investigate surface morphology, surface roughness, and determine the grain size of the titanium oxide thin films.
- Raman spectroscopy was used to analyze the phase composition of titanium oxide, based on measurements taken from the silicon substrate.
- Silver (Ag) and gold/silver (Au/Ag) electrodes were deposited onto the titanium oxide surface via the thermal evaporation.
- The first XPS measurement was performed on the as-deposited silver surface.
- A two-probe measurement system was used for current–voltage characterization.
- The fabricated samples were stored in a desiccator for 8 weeks.
- XPS measurements were repeated on the silver surface after 8 weeks to evaluate oxidation tendencies and surface chemistry.
- The long-term stability of the structures was assessed through endurance tests conducted 8 weeks after fabrication.
- The HRS/LRS ratios of structures with varying titanium oxide thicknesses were calculated.
- Eight weeks after fabrication, boundary layers and the elemental distribution of Ag/TiO_x(25 nm)/ITO were examined in detail using FIB-STEM-EDX.

3 Results and discussions

Titanium oxide thin films deposited on ITO substrates with varying thicknesses using the sputtering method were initially analyzed through SEM–EDX and AFM. SEM–EDX was employed to examine the distribution and concentration of elements on the surface. SEM–EDX mapping assessed the uniformity of titanium oxide deposition, contributing to determining thin film thickness and evaluating the effectiveness of the deposition process. AFM was used to analyze the surface morphology, roughness, and average grain size of thin films. AFM provided high-precision surface images and offered detailed insights into the surface structures of thin films. Raman spectroscopy was utilized to study crystal structures and material phases as it provides essential information on the structural properties of TiO_x thin films. To observe the oxidation state of the silver top electrodes, XPS measurements were performed twice, once after deposition was completed and again after 8 weeks. These periodic XPS analyses provided insights into the relation between electrode surface oxidation and device endurance and performance over a certain time frame. Samples were prepared using FIB to analyze the layers using STEM-EDX on cross-section. While STEM visualized the boundaries between layers, EDX mapping was used to analyze the chemical distribution of elements within the memristor layers. A two-probe measurement system was used to analyze the electrical behavior of TiO_x films with 25, 50, and 100 nm thicknesses.

3.1 SEM–EDX

Figure 2 presents the SEM–EDX analysis results of titanium oxide thin films with different thicknesses deposited on ITO-coated glass substrates by sputtering. The first image in each row shows the SEM image on the surface of the thin film. Figure 2 shows the EDX mapping results, where the green, blue, and purple colors correspond to titanium, oxygen, and indium, respectively. The distribution and density of elements in the EDX maps of films with varying thicknesses differ. As the thickness of the coating increases, the density of titanium and oxygen in the deposited area also increases. Additionally, there is a noticeable color transition in the area between the coated and uncoated surfaces.

The line scan analysis results of titanium oxide thin films deposited in different thicknesses on ITO-coated glass substrate by sputtering method are presented in Fig. 3. Line scan analyses demonstrate the element density on a particular line. This method is quite useful to examine how the elements on the surface of the film are distributed along the horizontal axis and how their distribution within the substrate changes. The change in the density of the titanium signal emphasizes the thickness profile of the coating. Consequently, in this analysis, film thickness and composition differences were monitored through the profile of the titanium signal.

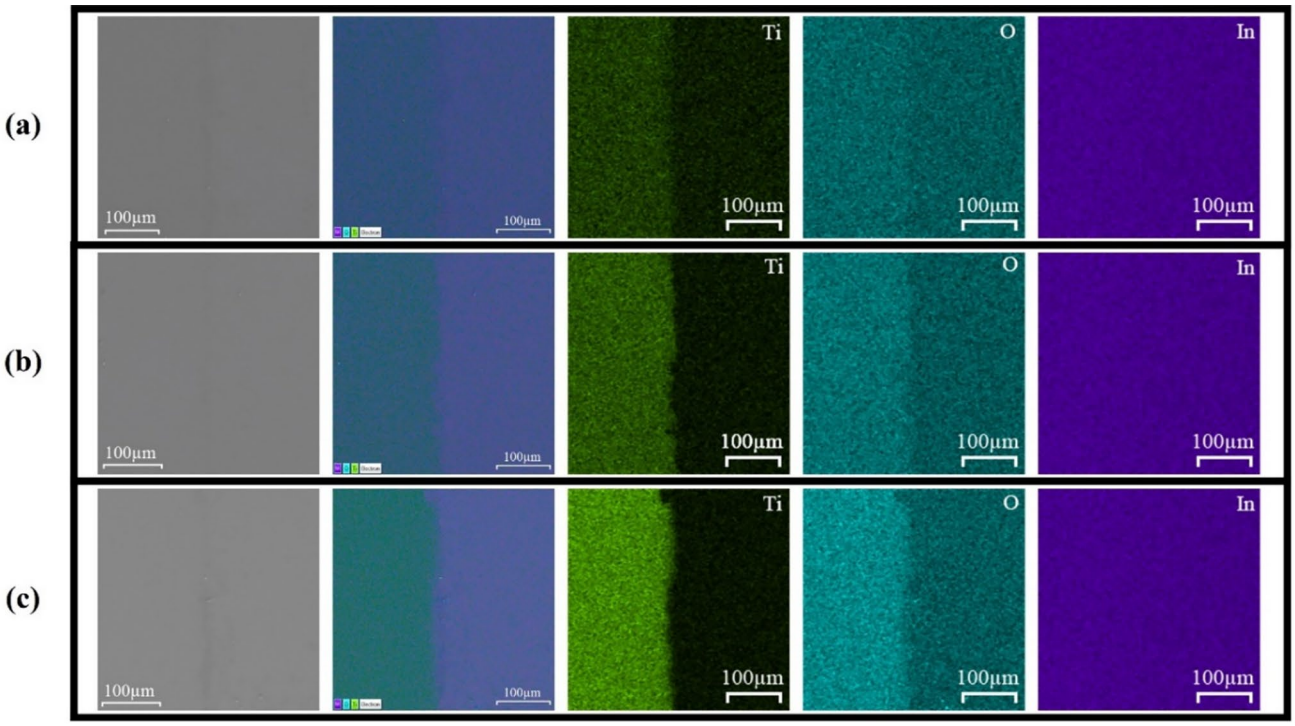


Fig. 2 SEM–EDX elemental mapping images of titanium oxide (TiO_x) thin films deposited on ITO-coated glass substrates by RF magnetron sputtering. The images correspond to films with thicknesses of **a** 25 nm, **b** 50 nm, and **c** 100 nm, respectively

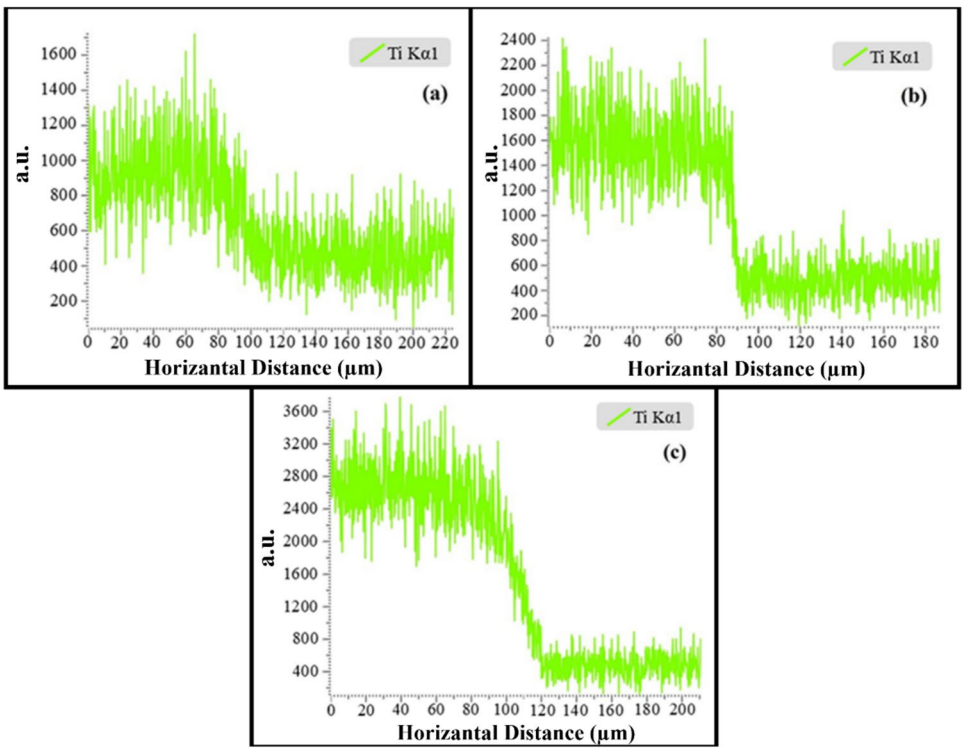


Fig. 3 EDX line scan analysis of Ti Kα1 signal across titanium oxide (TiO_x) thin films deposited on ITO-coated glass substrates via RF magnetron sputtering. The line scans were obtained for films with thicknesses of **a** 25 nm, **b** 50 nm, and **c** 100 nm, respectively. The signal intensity along the horizontal direction confirms the lateral distribution of Ti across the film boundary. Higher signal intensity corresponds to the TiO_x region, while a sharp drop indicates the transition to the undeposited ITO region

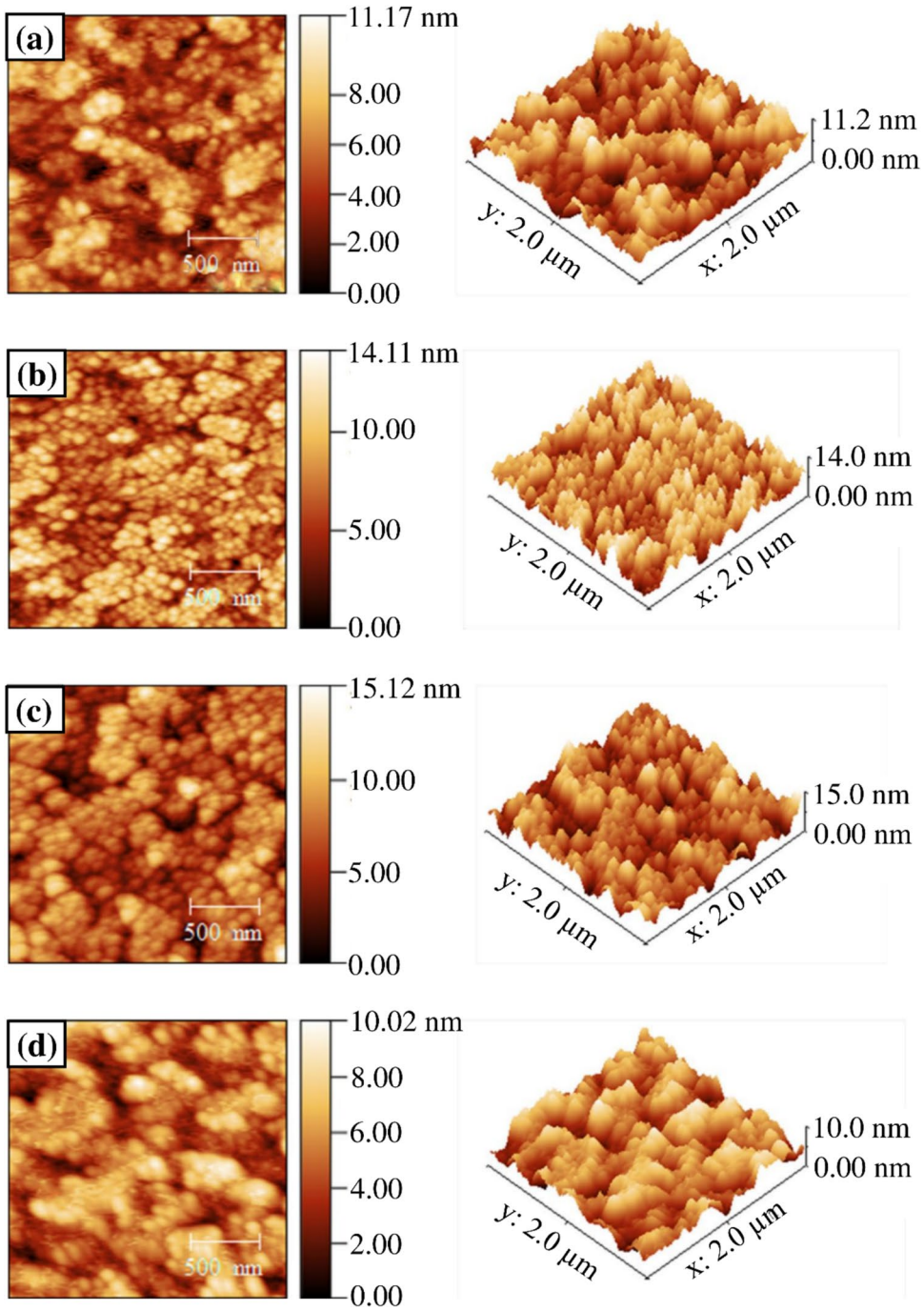


Fig. 4 2D and 3D AFM topography images of **a** bare ITO-coated glass substrate and titanium oxide thin films sputtered onto ITO with thicknesses of **b** 25 nm, **c** 50 nm, and **d** 100 nm. All images were acquired in tapping mode in a scan area of $2 \times 2\ \mu m^2$

3.2 AFM

AFM analysis examines the surface morphology, roughness, and surface structure of thin films. Figure 4 presents 2D and 3D AFM images of titanium oxide thin films with varying thickness deposited on ITO-coated glass substrates. Gwyddion software was used to analyze AFM images [43]. The 2D images visually represent the general structure, and surface roughness of the thin films, accompanied by a color bar. The maximum height on the scale indicates the highest peak on the surface. 3D images display the height distribution of the surface in more detail. Through these images, the roughness and grain of the film surface can be analyzed more clearly.

Table 1 Surface roughness and average grain size values obtained from AFM analysis of titanium oxide thin films with different thicknesses (25–50-100 nm), corresponding to the topography images presented in Fig. 4. Results show that surface roughness increases with film thickness, indicating enhanced grain growth in thicker films

Thickness	RMS roughness (nm)	Average grain size (nm)
25 nm	1.69	60
50 nm	2.08	80
100 nm	2.15	90

Each representative image displays a different average surface roughness, which demonstrates the impact of the deposition process on the surface. The morphological parameters of titanium oxide thin films, based on thickness, are summarized in Table 1. In AFM analysis, the root mean square (RMS) value is widely used to characterize surface roughness. RMS is calculated by averaging the squares of all height data on the surface, providing a statistical measure that effectively represents the magnitude of surface hills and pits. For titanium oxide thin films, the roughness value showed a limited increase with thickness. At 50 nm, a notable rise in roughness was observed compared to the 25 nm thickness, while at 100 nm, the roughness value remained relatively stable. This reveals that the surface morphology of titanium oxide films may be less sensitive to thickness change [44]. Also, the grain size of titanium oxide thin films increased with thickness. The grain size grew from 60 nm at 25 nm thickness to 80 nm and 90 nm at 50 nm and 100 nm thickness, respectively. This indicates a tendency for grain growth as the thickness of the titanium oxide thin films increases [45, 46].

In summary, the comparison of titanium oxide thin films with different thicknesses revealed that the highest roughness value and grain size were observed in the 100 nm film, as shown in Fig. 4d. The RMS surface roughness values increased from 1.69 nm to 2.15 nm with increasing film thickness. AFM analysis of titanium oxide thin films demonstrated that roughness values followed a trend dependent on coating thickness.

3.3 Raman spectrum

Raman spectral analysis was performed on titanium oxide thin films deposited on ITO-coated glass substrates using the sputtering method. However, the use of the ITO substrate resulted in interference, as the signal from the substrate overlapped with the Raman signal of the thin film, preventing the acquisition of clear signals. This spectral interference complicated the detection of characteristic Raman peaks of titanium oxide. This situation poses challenges, especially in thin films or when the Raman signal of the material is weak. Alternative substrates, like silicon substrates, are preferred to avoid such spectral interferences. The Raman spectrum of silicon substrate has a distinct peak at approximately 520 cm^{-1} [47, 48]. This peak does not cause interference in the spectrum, as it typically does not overlap with the spectral region of the analyzed thin films. Consequently, Raman measurements were conducted on a 100 nm titanium oxide thin film deposited on a silicon (100) substrate (Microchemicals). Titanium oxide exhibits three primary crystalline phases rutile, anatase, and brookite, each possessing distinct structural and electronic properties [49]. As shown in Fig. 5, the Raman peaks observed at 196.83 cm^{-1} and 607.43 cm^{-1} for the titanium oxide thin film on the silicon substrate correspond to the anatase and rutile phases of titanium oxide, respectively [50–53].

3.4 XPS

XPS analysis was utilized to investigate the changes in the surface chemistry of silver contact electrodes deposited using thermal evaporation. This technique is highly significant for examining the chemical structure and composition of materials. All samples fabricated were stored in a desiccator. The oxidation of the silver contact electrode was investigated by comparing XPS measurements taken at 8-week intervals from the sample stored in the desiccator. XPS analysis offered valuable insights into the performance and endurance properties of silver contacts by accurately revealing the oxidation processes occurring on the surface and their effects on the chemical composition. The XPS analysis results are presented in Fig. 6 and Fig. 7, with the graphs generated after baseline correction was applied to the raw data.

The XPS spectra obtained from the surface of a 150 nm silver thin film deposited by thermal evaporation

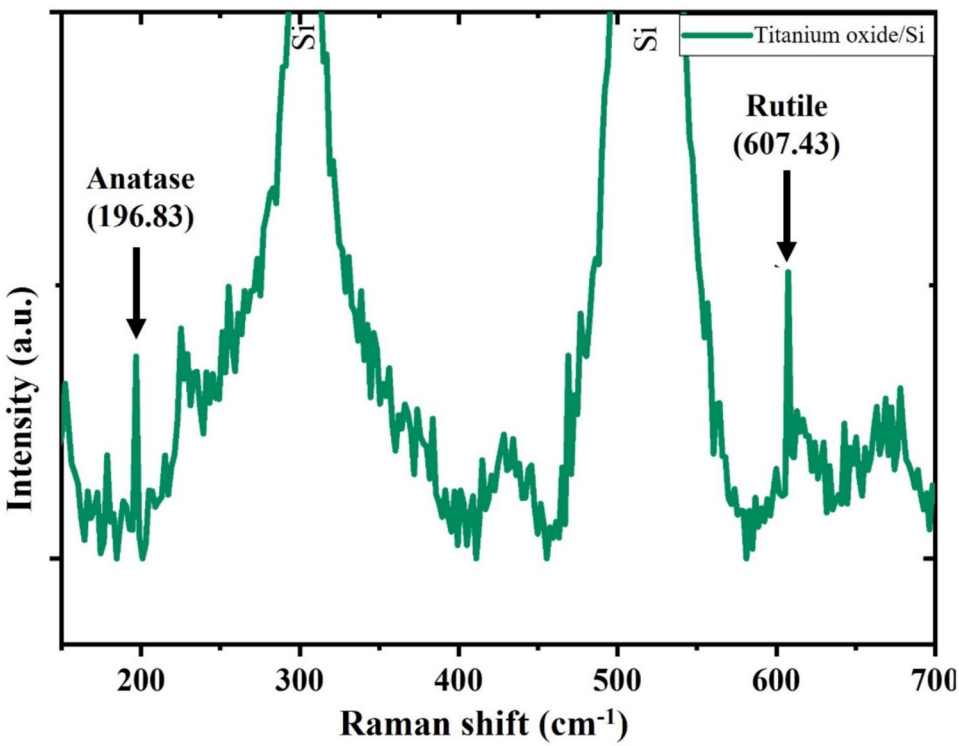


Fig. 5 Raman spectrum of a titanium oxide thin film with a thickness of 100 nm deposited on a Si substrate by RF magnetron sputtering. The measurement was performed at room temperature using a 785 nm excitation laser. The characteristic Raman peak observed at 196.83 cm^{-1} corresponds to the anatase phase of titanium oxide, while the peak at 607.43 cm^{-1} indicates the presence of the rutile phase. Peaks attributed to the Si substrate are also visible around ~ 300 cm^{-1} and ~ 520 cm^{-1}

are presented in Fig. 6. The full XPS scan of the surface is given in Fig. 6a, while the deconvolutions of the 3d and 1s spectra of silver are given in Fig. 6b and c, respectively. In the deconvolution of the 3d spectrum of silver, the peaks shown in red correspond to metallic silver, indicating its oxidation states and binding energies. The Ag $3d_{5/2}$ and Ag $3d_{3/2}$ peaks originate from the 3d orbital of Ag [54, 55]. The Ag $3d_{5/2}$ peak is located at a lower energy level, while the Ag $3d_{3/2}$ peak is located at a higher energy level, and both peaks belong to the 3d orbital of Ag (metallic silver).

Metallic silver peaks were observed at energies of 368.6 eV (Ag $3d_{5/2}$) and 374.6 eV (Ag $3d_{3/2}$). These values demonstrate that silver is in metallic form [56, 57]. The peaks introduced in blue shades illustrate Ag(I) oxidation, meaning the presence of Ag^+ ions. This indicates the Ag(I) state, which is the oxidized state of silver. The Ag(I) peaks were observed at energy values of 367.8 eV ($3d_{5/2}$) and 373.8 eV ($3d_{3/2}$). These peaks indicate the presence of silver ions (Ag^+) and represent the state of Ag(I), which is the oxidized state of silver [58, 59]. The results reveal that metallic silver and oxidized (Ag^+) silver are on the surface. The deconvolution process separated these two different chemical states of silver, providing a detailed analysis of the chemical composition of the silver on the surface. The deconvolution of the O 1s spectrum is typically performed to reveal the different chemical bonds and states of oxygen on the surface. In Fig. 6c, the peaks observed in the O 1s spectrum vary depending on the different bonding states of oxygen atoms. Generally, the peaks at 530.0 eV, 530.9 eV, and 532.6 eV are associated with O^{2-} species in the cage, oxygen vacancies or defects, and chemically absorbed or dissociated oxygen species [60].

In Fig. 7, XPS spectra taken 8 weeks after the production of the silver contact electrodes are presented. While the full scan of the surface is provided in Fig. 7a, the deconvolutions of the silver and oxygen spectra are correspondingly presented in Fig. 7b and c. Full scan spectra are obtained over a wide binding energy range to detect the elements on the surface. When the XPS spectra taken from the silver contact points after 8 weeks are examined, the increase in the C peak suggests that changes such as surface contamination or reactions with other elements may have occurred [61]. In Fig. 7a, metallic silver peaks are observed at 369.4 eV (Ag $3d_{5/2}$) and 375.3 eV (Ag $3d_{3/2}$) [56, 57]. The deconvolution of the Ag 3d and O 1s binding energy regions was performed to provide a more detailed interpretation of the spectrum. Comparing Fig. 6b and Fig. 7b, it is observed that

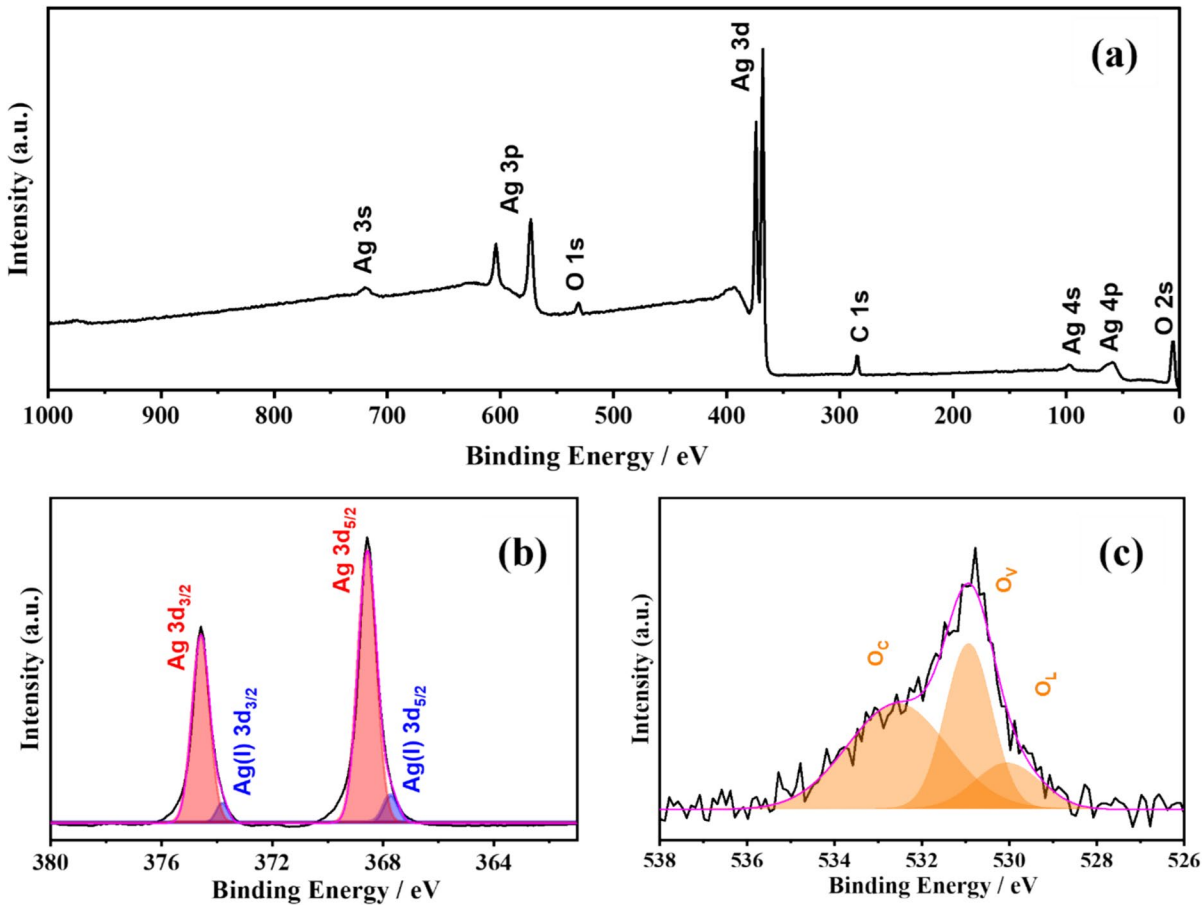


Fig. 6 XPS spectra of as-deposited silver contacts deposited on titanium oxide thin films using thermal evaporation method **a** full-scan spectrum showing the presence of Ag, O, and C elements,**b** High-resolution deconvoluted Ag 3d spectrum exhibiting double peaks, corresponding to Ag $3d_{3/2}$ and Ag $3d_{5/2}$, respectively, indicating metallic silver, and **c** High-resolution deconvoluted O 1s spectrum

there is a shift in the Ag (I) peak positions ($3d_{5/2}$ at 368.5 eV and $3d_{3/2}$ at 374.5 eV), as well as an increase in the intensity of the Ag (I) peaks. This indicates that the silver oxidized, and a layer resembling silver oxide formed on the surface. In Fig. 7c, the peaks at 530.6 eV, 531.4 eV, and 532.6 eV can be associated with O^{2-} species in the cage, oxygen vacancies or defects, and chemically absorbed or dissociated oxygen species [62]. Also, C peaks may rise due to carbon contamination.

3.5 STEM-EDX

In this study, a detailed analysis of the layers of the Ag/TiO_x(25nm)/ITO memristor structure was performed using FIB and STEM-EDX techniques. Initially, high-precision cross-sections of the sample were prepared using the FIB method to examine the interlayer properties of the memristor structure. The preparation of these sections allowed for a clear observation of nanometer-sized layers. The sections obtained by the FIB method were then subjected to a detailed structural examination on the STEM device. STEM provides high-resolution imaging of the boundaries between the sample layers. In addition, EDX mapping was performed for elemental analysis of layers, which is presented in Fig. 8. EDX mapping was utilized to determine the distribution of elements in the layers making up the memristor and to examine the composition of these layers in detail. As a result of EDX mapping, the distribution of the elements in the Ag, Ti, and In layers were determined, and possible element transitions between layers were identified. In addition, no significant traces of

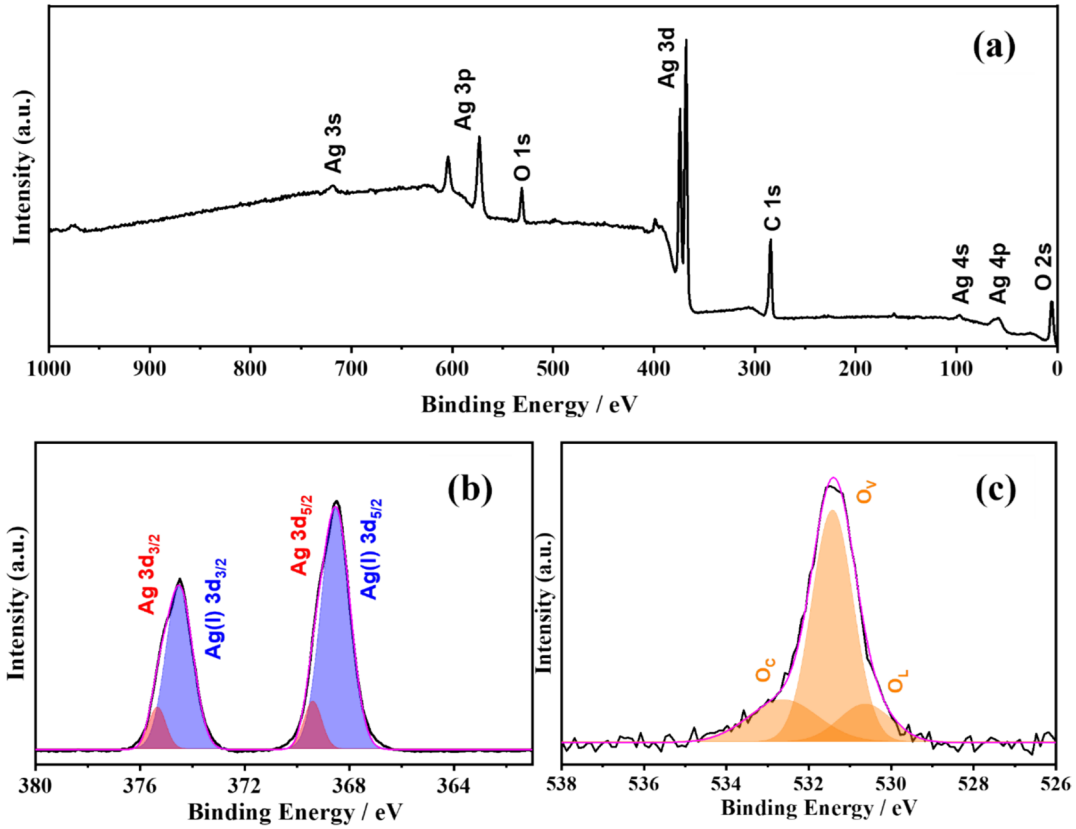


Fig. 7 XPS spectra of thermally evaporated silver contacts after 8 weeks of storage in a desiccator at room temperature **a** Full-scan spectrum confirming the presence of Ag, O, and C elements on the surface, **b** High-resolution deconvoluted Ag(I) 3d spectrum exhibiting double peaks, corresponding to Ag(I) $3d_{3/2}$ and Ag(I) $3d_{5/2}$, respectively, indicating oxidized silver Ag(I), **c** High-resolution deconvoluted O 1s spectrum

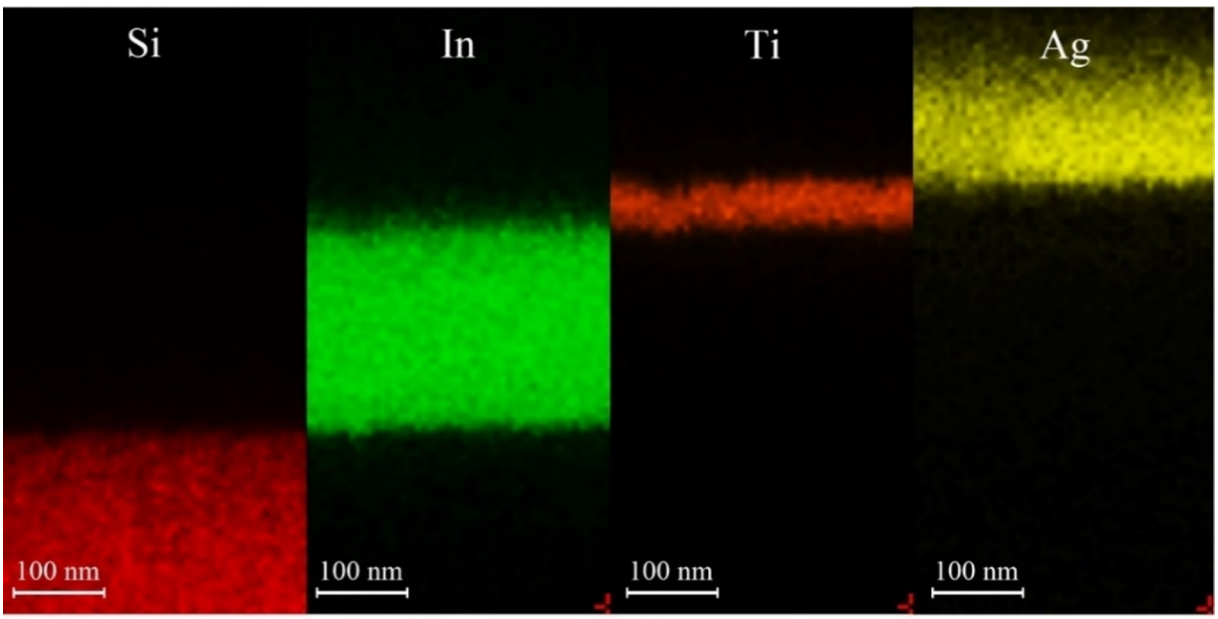


Fig. 8 Cross-sectional STEM-EDX elemental mapping of the Ag/TiO_x(25 nm)/ITO device, acquired from a sample prepared using a focused ion beam (FIB), after being stored in a desiccator for 8 weeks

diffusion were observed between the silver and titanium oxide active layers. Since ITO-coated glass is used as the substrate, there are silicon and indium layers, followed by titanium and a silver layer at the top. The mapping shows a clear transition between the layers, with a uniform distribution of elements, which is a factor that may positively influence the performance of the memristor.

3.6 Electrical characterization

The memristive properties of Ag/TiO_x/ITO and Au/Ag/TiO_x/ITO structures, obtained from titanium oxide thin films with different thicknesses deposited on ITO substrates using the sputtering method, were revealed using a two-terminal current–voltage measurement system. Ag and Ag/Au were used as the top and ITO as the bottom electrodes. Measurements were conducted by applying voltage to both the top and bottom electrodes, followed by a performance evaluation for both Ag and Ag/Au configurations. ITO was selected as the bottom electrode due to its conductivity and ability to simplify fabrication. Figure 9 compares the current–voltage characteristics at the 1st and 1000th cycles of memristor structures with three different titanium oxide layer thicknesses (25 nm, 50 nm, 100 nm) for Ag and Au/Ag top electrodes. These measurements were performed within 7 days of fabricating the memristor structures. The blue curves represent the 1st cycle, while the red curves depict the 1000th cycle. The characteristic current–voltage curves obtained over 1000 cycles were prepared to examine the electrical properties and cyclic stability of sandwich-type memristor structures with different titanium oxide layer thicknesses. For the Ag/TiO_x/ITO structure with 25 nm titanium oxide layer, the voltage was applied between -3 V and 3 V, and the corresponding current responses were recorded. Similarly, for the Ag/TiO_x/ITO and Au/Ag/TiO_x/ITO structures with 50 nm and 100 nm titanium oxide layers, voltage sweeps were performed between -2 V and 2 V, and current–voltage measurements were recorded. Although the top electrode of each memristor was different, they shared the same node because their bottom electrodes were short-circuited, as previously depicted in Fig. 1. The performance of the memristors was evaluated based on I-V changes across cycles, comparing the current–voltage measurements from the 1st and 1000th cycles for each structure. Current–voltage measurements from the 1st and 1000th cycles were compared for each structure.

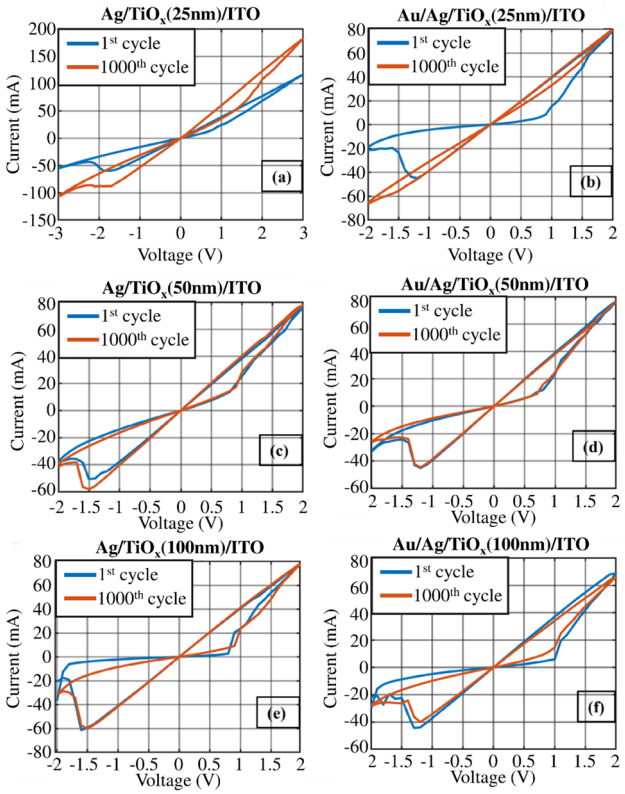


Fig. 9 I–V measurement of as-fabricated memristor structures based on TiO_x thin films with different top electrode configurations and thicknesses; **a** Ag/TiO_x(25 nm)/ITO, **b** Au/Ag/TiO_x(25 nm)/ITO, **c** Ag/TiO_x(50 nm)/ITO, **d** Au/Ag/TiO_x(50 nm)/ITO, **e** Ag/TiO_x(100 nm)/ITO, and **f** Au/Ag/TiO_x(100 nm)/ITO. All measurements were carried out under room conditions using a state-of-the-art measurement system with a voltage sweep for consecutive 1000 cycles (step size: 100 mV) The 1st and 1000th I–V curves show typical memristive behavior with clear hysteresis loops, indicating resistive switching characteristics

This analysis is crucial for evaluating the long-term memory and endurance characteristics of memristors. Maintaining a broad hysteresis curve for memristors is vital, as it offers a more pronounced memory and distinguishable resistance ratio. However, the narrowing of the hysteresis curve observed during the 1000th cycle in Fig. 9b reveals performance degradation over time, leading to a decline in initial memory characteristics. The structure with thickness of a 50 nm operates at lower currents than the 25 nm structure, and at the 1000th cycle, its current–voltage curve exhibits a wider and more stable behavior. This indicates that 50 nm provides more cyclic stability than 25 nm.

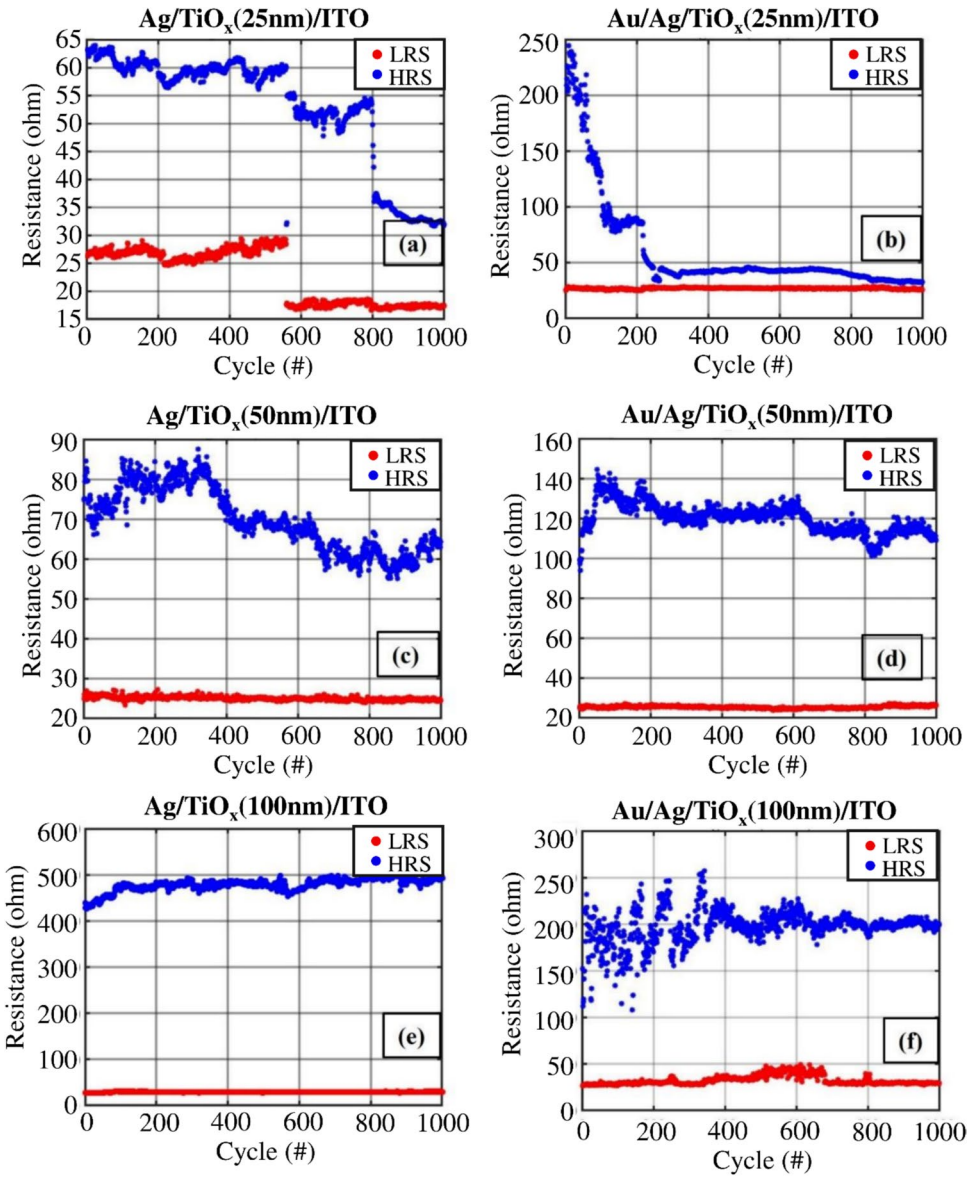


Fig. 10 Endurance performance of memristor structures stored in a desiccator for 8 weeks, with different TiO_x thicknesses and electrode configurations; **a** Ag/TiO_x(25nm)/ITO, **b** Au/Ag/TiO_x(25nm)/ITO, **c** Ag/TiO_x(50nm)/ITO, **d** Au/Ag/TiO_x(50nm)/ITO, **e** Ag/TiO_x(100nm)/ITO, and **f** Au/Ag/TiO_x(100nm)/ITO. The test involved applying – 1.5 V SET and 1.5 V RESET voltages and measuring the HRS and LRS values at 0.5 V

Eight weeks after the fabrication, the titanium oxide-based memristor structures exhibiting bipolar resistive switching behavior during I-V measurements over 1000 cycles were tested for endurance over an additional 1000 cycles (Fig. 10). The test involved applying -1.5 V SET and 1.5 V RESET voltages and measuring the HRS and LRS values at 0.5 V read voltage using a state-of-the-art measurement system for 1000 cycles. Table 2 presents the average HRS/LRS ratios measured at 0.5 V read voltage for titanium oxide-based memristor structures with varying thicknesses and two different top electrode materials (Ag and Au/Ag).

Figure 11 depicts the HRS/LRS ratios of the investigated memristor structures with standard deviations given by error bars. The results reveal a significant increase in the HRS/LRS ratio with increasing thickness when the silver top electrode is used. Notably, at a thickness of 100 nm, this ratio reached 16.07, the highest value observed. This finding indicates that the thicker memristor active layer enhances switching performance and more distinguishable resistance states between the ON and OFF states. Furthermore, the higher HRS/LRS ratio suggests that this structure could potentially store data more reliably. For memristors with Au/Ag electrodes, the HRS/LRS ratio also increases with thickness. At a thickness of 50 nm,

Table 2 Comparison of the average HRS/LRS resistance ratios of memristor structures incorporating titanium oxide layers of varying thicknesses (25–50-100 nm). Each device was subjected to 1000 endurance cycles using a read voltage of 0.5 V after 8 weeks of storage in a desiccator

Thickness	HRS/LRS ratio (Ag top electrode)	HRS/LRS ratio (Au/Ag top electrode)
25 nm	2.34	2.24
50 nm	2.78	4.73
100 nm	16.07	6.22

the HRS/LRS ratio is 4.73, while at 100 nm, this ratio increases to 6.22. Although switching performance improves with increasing thickness in Au/Ag electrodes, the extent of this improvement is more limited than the improvement observed in silver electrodes.

This study presents a comprehensive approach since it investigated the surface, morphological, structural, and electrical properties of titanium oxide thin films sputtered on ITO substrates with different thicknesses (25 nm, 50 nm, and 100 nm). By combining AFM, SEM–EDX, XPS, Raman, FIB, STEM-EDX, and current–voltage characterization techniques, a comprehensive understanding of various properties of the memristor was achieved. While AFM [63, 64], SEM [65, 66] and XPS [67, 68] techniques are widely used in the literature, SEM–EDX, STEM-EDX, and FIB techniques are less commonly used [66, 69]. The comprehensive analysis of morphological, structural, chemical, and surface properties underscores the thoroughness of this study. Combining SEM–EDX, STEM-EDX, FIB, and AFM techniques facilitated an in-depth examination of the structures. Compared to previous studies, which typically focus on a single aspect (e.g., surface properties, chemical structure, or crystal structure) [63, 70], this study provides a multi-faceted analysis that integrates multiple perspectives.

AFM analyses demonstrated that the surface roughness and grain size of titanium oxide thin films increased with thickness. The roughness values of the films with thicknesses of 25 nm, 50 nm, and 100 nm were measured as 1.69 nm, 2.08 nm, and 2.15 nm, respectively. Similarly, the grain sizes increased from 60 to 90 nm with increasing film thickness. These findings align with the trend of increasing grain size and surface roughness with film thickness, as reported in the literature [71–73]. SEM–EDX analyses revealed that the elemental distribution of the films remained uniform with increasing thickness while the densities of titanium and oxygen increased. EDX mappings showed that the green regions, representing titanium-rich areas, became more prominent as the film thickness increased. Furthermore, STEM-EDX results indicated limited interlayer diffusion in memristor structures, with distinct elemental boundaries forming between layers.

Current–voltage measurements taken when the structures were first produced (Fig. 9) showed changes in their electrical behavior depending on their thickness. XPS data collected after 8 weeks showed increased oxygen content on the silver contact surface. Despite this, the memristor structures exhibited bipolar resistive switching behavior in electrical measurements. This suggests that while surface oxidation may influence the properties of memristors, it does not significantly impair their switching performance in an 8-week period. This information has the potential to provide insight into the packaging time for memristors after fabrication.

Comparative analysis method has been commonly presented in previous studies [13, 74]. Table 3 presents a comparative overview of reported titanium oxide-based resistive switching devices [13, 41, 42, 66, 67,

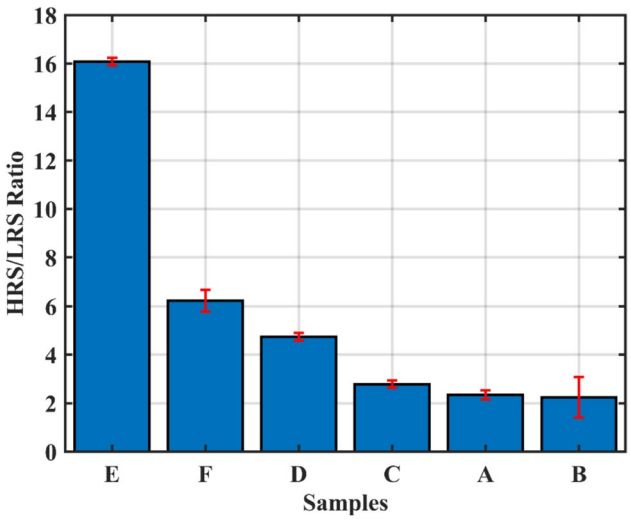


Fig. 11 Average HRS/LRS resistance ratios with standard deviations for memristor structures based on TiO_x thin films with varying thicknesses and electrode configurations; Sample **A** Ag/TiO_x(25 m)/ITO, **B** Au/Ag/TiO_x(25 nm)/ITO, **C** Ag/TiO_x(50 nm)/ITO, **D** Au/Ag/TiO_x(50 nm)/ITO, **E** Ag/TiO_x(100 nm)/ITO, **F** Au/Ag/TiO_x(100 nm)/ITO

Table 3 Comparison of titanium oxide-based memristive devices reported in recent literature, focusing on key structural, electrical, and characterization parameters. The table details device configurations, memristive layer thicknesses, deposition methods, HRS/LRS ratios, endurance cycles, and surface characterization techniques used in research

Device configuration	Active layer thickness	Active layer deposition/coating method	HRS/LRS ratio	Endurance cycle	Surface characterization techniques	References
Pt/TiO_x/Au	37 nm	PE-ALD	> 4.5*	2×10^3	XPS, Optical microscopy, HRTEM, FIB** STEM-EDS	[13]
Pt/TiO_x/Ti	40 nm	RF sputtering	7.96	10^2	TEM, XPS	[74]
Ti/Ir/TiO_x/p^{++}-Si	50 nm	PLD	10^2	10^2	SEM, XRD, AFM, XPS	[75]
Ag/TiO_x/ITO	392 nm	Sputtering	2.90	75	SEM, EDS	[41]
Ag/TiO_2/TiO_x/ITO	NA	Spin coating	2×10^3	10	HRTEM, EDS, XPS	[42]
Pt/TiO_x/Al	18 nm	RF sputtering	> 4*	630	Optical microscopy, AFM, TEM, FIB**, EDS, XPS	[66]
Pt/TiO_x/Pt	7.2/40/80 nm	PLD	2 (7.2 nm) 6 (40 nm) 1.4 (80 nm)	> 8	XRR, XPS, PL	[67]
Au/TiO_2 nanowire/FTO	1.5 µm	Hydrothermal	$> 10^2$	NA	XRD, SEM, XPS	[76]
Ag/TiO_2/FTO	7 µm	Hydrothermal	> 10	15	XRD, SEM, EDX, PL	[77]
Au/TiO_2/Au	10 nm	ALD	10^6	10^3	HRTEM, FIB**, SEM, Optical microscopy	[78]
Cu/N-TiO_2/FTO Al/N-TiO_2/FTO	NA	Hydrothermal	> 10	10^2	SEM	[79]
Au/Ti/TiO_2/Ti/FTO	50 nm	e-beam evaporation	20	2×10^2	XRD, XPS, AFM, UV–Vis, TEM, EDS	[80]
Ag/TiO_x/ITO Au/Ag/TiO_x/ITO	25/50/100 nm	RF sputtering	2.34 (25 nm) 2.78 (50 nm) 16.07 (100 nm) 2.24 (25 nm) 4.73 (50 nm) 6.22 (100 nm)	10^3	SEM–EDX, AFM, Raman, XPS, FIB**, STEM-EDX, Optical microscopy	This study

*Estimated from endurance measurements presented in the relevant references

**FIB is used for sample preparations in relevant references

74–80]. Various deposition techniques, including RF sputtering, DC sputtering, pulsed laser deposition (PLD), atomic layer deposition (ALD), and hydrothermal methods, have been employed across these studies, resulting in diverse film thicknesses and switching characteristics. The table highlights key aspects such as device configuration, switching layer thickness, fabrication methods, HRS/LRS ratios, endurance performance, and the characterization techniques used in each study. Our work offers a holistic and in-depth investigation by integrating some of the most advanced characterization techniques in the field. The present work provides the systematic exploration of thickness-dependent switching behavior and

the application of multiple advanced and rarely co-employed characterization techniques. Notably, the use of EDX line scan, STEM-EDX, and Raman spectroscopy along with other methods enables deeper insight into the structural and electrical dynamics of the device.

4 Conclusion

This research aims to understand the relationship between surface science, device durability, and memristor fabrication approaches. Here, the surface morphology and electrical properties of Ag/TiO_x/ITO and Au/Ag/TiO_x/ITO structures were examined based on the thickness of the titanium oxide active layer. Surface characterization techniques such as SEM and AFM were utilized to analyze the roughness, grain size, and surface structures of the films. Significant increases in surface roughness and particle size were observed with the increase in the thickness of the titanium oxide thin films. Increase in the thickness positively affected the performance of the memristor and was found to contribute to its long-term stable operation. The increase in thin film thickness significantly affected both electrical stability and the switching performance of the memristor. The I–V characteristics exhibit clear bipolar switching with well-defined SET and RESET voltages and a noticeable hysteresis loop, consistent with the formation and rupture of conductive filaments within the TiO_x layer. Notably, all devices demonstrated forming-free bipolar switching behavior, eliminating the need for high-voltage electroforming. When combined with the stable endurance performance demonstrated through 1000 consecutive SET/RESET cycles measured at ± 1.5 V with a read voltage of 0.5 V, these results underscore the reliability and practical applicability of the proposed memristor structures. The resistive switching behavior observed in the Ag/TiO_x/ITO and Au/Ag/TiO_x/ITO memristor structures is primarily attributed to a filamentary-type switching mechanism. Ag electrodes exhibited higher endurance in thicker samples (100 nm). According to experimental observations, gold-passivated silver electrodes also performed well but did not provide a resistance ratio as high as that of silver electrodes. The highest HRS/LRS ratio of 16.07 was achieved in the Ag/TiO_x/ITO structure with a 100 nm thick titanium oxide layer. As a result, the study concludes that both the electrode material and the thickness of the active layer play a critical role in titanium oxide-based memristors. The effects of different substrates, bottom electrode materials, and deposition parameters can be subject to further investigations in future work.

Acknowledgements

The authors would like to acknowledge the use of the services and facilities provided by the Gazi University Photonics Research Center, UNAM at Bilkent University, and HUNITEK at Hacettepe University. The authors would like to extend their special thanks to Mustafa Guler from UNAM for STEM measurements and Erhan Karaca for his support in XPS plots. This work is part of Kubra Saka's PhD studies.

Author contributions

The manuscript is written by KS and revised by DG. DG supervised the research activities. KS, DG, HIE, CB, and SO contributed to the design of experiments and surface analysis. The manuscript is approved by all authors.

Funding

Open access funding provided by the Scientific and Technological Research Council of Türkiye (TÜBİTAK). Open access funding provided by the Scientific and Technological Research Council of Türkiye (TÜBİTAK). The authors have not disclosed any funding.

Data availability

The data supporting the findings of this study are available upon reasonable request from the authors.

Declarations

Conflict of interest No potential conflict of interest was reported by the authors.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

1. L. Chua, IEEE Transactions on circuit theory **18**, 507 (1971)
2. D.B. Strukov, G.S. Snider, D.R. Stewart, R.S. Williams, Nature **453**, 80 (2008)
3. C. Weilenmann, A.N. Ziogas, T. Zellweger et al., Nat. Commun. **15**, 6898 (2024)
4. Y.H. Jang, W. Kim, J. Kim et al., Nat. Commun. **12**, 5727 (2021)
5. C. Yakopcic, M.Z. Alom, T.M. Taha, International joint conference on neural networks (IJCNN) IEEE (2017)
6. J.H. Yoon, Z. Wang, K.M. Kim et al., Nat. Commun. **9**, 417 (2018)
7. K. Sun, J. Chen, X. Yan, Adv. Func. Mater. **31**, 2006773 (2021)
8. M. Hu, H. Li, Y. Chen, Q. Wu, G.S. Rose, R.W. Linderman, IEEE transactions on neural networks and learning systems **25**, 1864 (2014)
9. J. Simmons, R. Verderber, Proc. R. Soc. London Ser. A **301**, 77 (1967)
10. D.-H. Kwon, K.M. Kim, J.H. Jang et al., Nat. Nanotechnol. **5**, 148 (2010)
11. E. Gul, D. Gokcen, ECS Journal of Solid State Science and Technology **9**, 054004 (2020)
12. D. Gokcen, O. Şentürk, E. Karaca, N.Ö. Pekmez, K. Pekmez, J. Mater. Sci. Mater. Electron. **30**, 5733 (2019)
13. F. Ullah, M. Tarkhan, Z. Fredj, Y. Su, T. Wang, M. Sawan, Nano Express **5**, 015003 (2023)
14. J. Qiu, A. Shih, W. Zhou, Z. Mi, I. Shih, J. Appl. Phys. (2011). https://doi.org/10.1063/1.3599952
15. K. Saka, D. Gokcen, J. Mater. Sci. Mater. Electron. **33**, 21811 (2022)
16. J.-W. Han, M. Meyyappan, AIP Adv. (2011). https://doi.org/10.1063/1.3645967
17. P. Zhou, M. Yin, H. Wan, H.-B. Lu, T.-A. Tang, Y.-Y. Lin, Appl. Phys. Lett. (2009). https://doi.org/10.1063/1.3078824
18. H. Jiang, L. Han, P. Lin et al., Sci. Rep. **6**, 28525 (2016)
19. H. Lee, Y. Chen, P. Chen et al., IEEE Electron Device Lett. **31**, 44 (2009)
20. N. Ilyas, J. Wang, C. Li et al., J. Mater. Sci. Technol. **97**, 254 (2022)
21. J. Yao, Z. Sun, L. Zhong, D. Natelson, J.M. Tour, Nano Lett. **10**, 4105 (2010)
22. J. Gibbons, W. Beadle, Solid-State Electron. **7**, 785 (1964)
23. Y. Li, P. Fang, X. Fan, Y. Pei, Semicond. Sci. Technol. **35**, 055004 (2020)
24. Y.C. Yang, F. Pan, F. Zeng, New J. Phys. **12**, 023008 (2010)
25. B.P.S. Rathore, R. Prakash, D. Kaur, Curr. Appl. Phys. **18**, 102 (2018)
26. Y.C. Yang, F. Pan, Q. Liu, M. Liu, F. Zeng, Nano Lett. **9**, 1636 (2009)
27. H. Xu, D.H. Kim, Z. Xiahou et al., J. Alloy. Compd. **658**, 806 (2016)
28. K. Krishnan, T. Tsuruoka, C. Mannequin, M. Aono, Adv. Mater. (Deerfield Beach, Fla.) **28**, 640 (2015)
29. Y. Han, I. Chung, S. Park, K. Cho, S. Kim, J. Nanosci. Nanotechnol. **13**, 6208 (2013)
30. K. Park, J.-S. Lee, Sci. Rep. **6**, 23069 (2016)
31. C.S. Dash, S. Sahoo, S. Prabaharan, Solid State Ionics **324**, 218 (2018)
32. J. Molina-Reyes, L. Hernandez-Martinez, Complexity **2017**, 8263904 (2017)
33. N.-N. Ge, C.-H. Gong, X.-C. Yuan, H.-Z. Zeng, X.-H. Wei, RSC Adv. **8**, 29499 (2018)
34. Y. Hu, D. Perello, M. Yun, D.-H. Kwon, M. Kim, Microelectron. Eng. **104**, 42 (2013)
35. I. Köymen, P.A. Göktürk, C. Kocabaş, Ş Süzer, Faraday Discuss. **213**, 231 (2019)
36. Y. Yang, P. Gao, S. Gaba, T. Chang, X. Pan, W. Lu, Nat. Commun. **3**, 732 (2012)
37. I. Valov, ChemElectroChem **1**, 26 (2014)
38. S. Hu, S. Wu, W. Jia et al., Nanosci. Nanotechnol. Lett. **6**, 729 (2014)

39. R. Waser, M. Aono, Nat. Mater. **6**, 833 (2007)
40. O. Kavehei, K. Cho, S. Lee, et al. IEEE 54th International Midwest Symposium on Circuits and Systems (MWSCAS) (2011)
41. H. Elshekh, H. Wang, S. Zhu, C. Yang, J. Wang, Chem. Phys. **580**, 112217 (2024)
42. M.H. Park, J.H. Jeong, W. Kim et al., J. Mater. Chem. C **12**, 6881 (2024)
43. D. Nečas, P. Klapetek, Open Phys. **10**, 181 (2012)
44. C.H. Heo, S.-B. Lee, J.-H. Boo, Thin Solid Films **475**, 183 (2005)
45. I.J. Badovinac, R. Peter, A. Omerzu et al., Thin Solid Films **709**, 138215 (2020)
46. K. Płacheta, A. Kot, J. Banas-Gac et al., Appl. Surf. Sci. **608**, 155046 (2023)
47. A.I. Pérez-Jiménez, D. Lyu, Z. Lu, G. Liu, B. Ren, Chem. Sci. **11**, 4563 (2020)
48. C. Smit, R. Van Swaaij, H. Donker, A. Petit, W. Kessels, M. Van de Sanden, J. Appl. Phys. **94**, 3582 (2003)
49. Y. Zhang, X. Xu, ACS Omega **5**, 15344 (2020)
50. C.-K. Chung, M. Liao, C. Lai, Thin Solid Films **518**, 1415 (2009)
51. Y. Ma, J.-B. Qiu, Y.-A. Cao, Z.-S. Guan, J.-N. Yao, Chemosphere **44**, 1087 (2001)
52. O. Frank, M. Zukalova, B. Laskova, J. Kürti, J. Koltai, L. Kavan, Phys. Chem. Chem. Phys. **14**, 14567 (2012)
53. V. Naik, D. Haddad, R. Naik, J. Benci, G. Auner, Mat. Res. Soc. Symp. Proc., **755**, (2003)
54. M. Al-Kuhaili, J. Phys. D Appl. Phys. **40**, 2847 (2007)
55. D. Lützenkirchen-Hecht, H.H. Strehblow, Surf. Interface Anal. **41**, 10, 820 (2009)
56. A.A. Ismail, Microporous Mesoporous Mater. **149**, 69 (2012)
57. Y.L. Cha, J.-H. Jo, D.-J. Kim, S.H. Kim, Nanomaterials **12**, 2060 (2022)
58. H. Karimi-Maleh, B.G. Kumar, S. Rajendran et al., J. Mol. Liq. **314**, 113588 (2020)
59. L. Tian, Z. Huang, X. Lu et al., Inorg. Chem. **62**, 1659 (2023)
60. https://sites.cardiff.ac.uk/xpsaccess/reference/oxygen, Accessed 6 February 2025
61. P. Bazant, I. Kuritka, L. Munster, L. Kalina, Cellulose **22**, 1275 (2015)
62. T.C. Kaspar, T. Droubay, S.A. Chambers, P.S. Bagus, J. Phys. Chem. C **114**, 21562 (2010)
63. Y. Khrapovitskaya, N. Maslova, I. Sokolov, Y. Grishchenko, D. Mamichev, M. Zanaveskin, Phys. Status Solidi C **12**, 202 (2015)
64. C. Lee, I. Kim, W. Choi, H. Shin, J. Cho, Langmuir **25**, 4274 (2009)
65. V. Shetty, S.M. Shanbogh, P. Anjaneyulu, K. Deepak, J. Mater. Sci. Mater. Electron. **35**, 1 (2024)
66. Z. Ye, M. Wu, P. Xu et al., J. Mater. Sci. Mater. Electron. **35**, 1223 (2024)
67. B. Özkal, S. Kazan, Ö. Karataş, G. Ekinci, L. Arda, B.Z. Rameev, Mater. Res. Exp. **10**, 125901 (2023)
68. K.P. Biju, X. Liu, E.M. Bourim et al., J. Phys. D Appl. Phys. **43**, 495104 (2010)
69. S. Srivastava, J.P. Thomas, N.F. Heinig, K. Leung, ACS Appl. Mater. Interfaces **9**, 36989 (2017)
70. V. Senthilkumar, A. Kathalingam, V. Kannan, J.-K. Rhee, Microelectron. Eng. **98**, 97 (2012)
71. A. Taherniya, D. Raoufi, Mater. Res. Exp. **6**, 016417 (2018)
72. I.K. Piltaver, R. Peter, I. Šarić et al., Appl. Surf. Sci. **419**, 564 (2017)
73. M.F. Hossain, M.S. Pervez, M. Nahid, Emerg. Mater. Res. **9**, 186 (2020)
74. D. Hasina, A. Mandal, S.K. Srivastava, A. Mitra, T. Som, Small 2408369 (2025)
75. D. Ju, S. Kim, Resul. Phys. **61**, 107796 (2024)
76. Z. Yu, T. Sun, B. Liu et al., J. Alloy. Compd. **858**, 157749 (2021)
77. M.S. Irshad, A. Abbas, H.H. Qazi et al., Mater. Res. Exp. **6**, 076311 (2019)
78. J. Lee, K. Song, K.-Y. Shin, W. Lee, Mater. Sci. Eng. B **314**, 117986 (2025)
79. Y. Yu, Z. Ding, Y. Ren, X. Wang, H. Quan, H. Jia, C. Jiang, ACS Omega **23**, 24601 (2024)
80. M.U. Rashid, S.A. Khan, F. Ghafoor et al., Curr. Appl. Phys. **64**, 16 (2024)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.